

SKKU 2024 Challenge: Dicke State Preparation

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Exercise 1

Quantum phase estimation (QPE) can be used to measure the Hamming weight of a quantum state and prepare Dicke states.

a) Write down the Hamiltonian whose eigenvalues can distinguish the Hamming weight of an n qubit quantum state.

The Hamiltonian whose eigenvalues differs by Hamming weight in n -qubit system can be constructed by the projector onto the space of fixed Hamming weight i . That is,

$$\mathcal{H} = \sum_{i=0}^{2^n-1} e_i P_i$$

We can choose $e_i = i$ where the eigenvalue becomes the Hamming weight i . Note that the given hamiltonian can have ${}_nC_i$ for e_i .

b) For an n qubit state, what is the minimum number of bits needed to measure the hamming weight of the state? Hint: The Hamming weight of the state is in the range $\{0, 1, \dots, n\}$

Let k denote the minimum number of bits required to measure hamming weight of the state. Then

$$k = \lceil \log_2(n+1) \rceil$$

where $\lceil a \rceil$ denotes the smallest integer larger than a .

Exercise 2

Using qiskit, create a function that takes in a quantum state in vector format as input and returns a quantum circuit that measures the Hamming weight of the state. Hint: see ref [1]. You can use qiskit's initialize module to initialize the states.

From the given reference, in order to measure Hamming weight of n qubit system, it requires k qubits as a control register as discussed in the Exercise 1 b). Let $|\Psi\rangle$ denote the wavefunction for whole system including control registers and prepared state. Given arbitrary state $|\psi\rangle$, the following algorithm starts with preparing $|\psi\rangle \otimes |0\rangle_C^{\otimes k}$. Then, act on the control register qubits with k Hadamard gates, we have:

$$|\Psi\rangle = |\psi\rangle \otimes H^{\otimes k} |0\rangle_C^{\otimes k} \quad (1)$$

$$= |\psi\rangle \otimes |+\rangle_C^{\otimes k} \quad (2)$$

Next, acting on $R_z(\gamma_j)_{C_j}$ gate on j qubit, labeled C_j where $\gamma_j = \frac{n\pi}{N+1} 2^{j-1}$.

$$|\Psi\rangle = |\psi\rangle \otimes \Pi_{j=1}^k R_z(\gamma_j)_{C_j} |+\rangle_{C_j} \quad (3)$$

Applying the following controlled-Z gate,

$$U = \Pi_{j=1}^k \left[|0\rangle\langle 0|_{C_j} \otimes I_S + |1\rangle\langle 1|_{C_j} \otimes R_z^{\otimes n}(\beta_j)_S \right] \quad (4)$$

where $\beta_j = \frac{2\pi}{N+1} 2^{j-1}$. Finally, IQFT (inverse quantum Fourier Transform) on register qubits and measure register qubits in the computational basis $\{|0\rangle\langle 0|, |1\rangle\langle 1|, \dots, |N\rangle\langle N|\}$.

Let me show a detailed process that how the state changes that enables measurement of Hamming weight. After the Eq. 3, the state becomes:

$$\begin{aligned} |y\rangle_C &\rightarrow \bigotimes_{j=1}^k R_z(\gamma_j)_{C_j} |y_k\rangle_{C_k} \dots |y_1\rangle_{C_1} \\ &= \exp \left\{ \left(\sum_{j=1}^k \frac{i\gamma_j}{2} (2y_j - 1) \right) \right\} |y_k\rangle_{C_k} \dots |y_1\rangle_{C_1} \\ &= \exp \left(\sum_{j=1}^k i\gamma_j y_j - \frac{i}{2} \sum_{j=1}^k \gamma_j \right) |y_k\rangle_{C_k} \dots |y_1\rangle_{C_1} \\ &= \exp \left(i\vec{\gamma} \cdot \vec{y} - \frac{i}{2} \Gamma \right) |y\rangle_C, \end{aligned} \quad (5)$$

where $\Gamma = \sum_{j=1}^k \gamma_j$. Thus,

$$|\Psi\rangle = |\psi\rangle \otimes \frac{1}{\sqrt{N+1}} \sum_{y=0}^N \exp \left\{ \left(i\vec{\gamma} \cdot \vec{y} - \frac{i}{2} \Gamma \right) \right\} |y\rangle_C \quad (6)$$

Next, considering the action of controlled-Z gate in Eq. 4,

$$\begin{aligned}
|y\rangle_C |\psi\rangle_S &\rightarrow |y\rangle_C R_z^{\otimes n}(y_1\beta_1) \cdots R_z^{\otimes n}(y_k\beta_k) |\psi\rangle_S \\
&= |y\rangle_C R_z^{\otimes n}(y_1\beta_1 + \cdots + y_k\beta_k) |\psi\rangle_S \\
&= |y\rangle_C R_z^{\otimes n}\left(\sum_{j=1}^k y_j\beta_j\right) |\psi\rangle_S \\
&= |y\rangle_C R_z^{\otimes n}(\vec{y} \cdot \vec{\beta}) |\psi\rangle_S,
\end{aligned} \tag{7}$$

thus, we have:

$$|\Psi\rangle = R_z^{\otimes n}(\vec{y} \cdot \vec{\beta}) |\psi\rangle_S \otimes \frac{1}{\sqrt{N+1}} \sum_{y=0}^N \exp\left\{\left(i\vec{\gamma} \cdot \vec{y} - \frac{i}{2}\Gamma\right)\right\} |y\rangle_C \tag{8}$$

Here,

$$\vec{\gamma} \cdot \vec{y} = \sum_{j=1}^k \gamma_j y_j = \frac{n\pi}{N+1} \sum_{j=1}^k y_j 2^{j-1} = \frac{n\pi y}{N+1}, \tag{9}$$

$$\vec{\beta} \cdot \vec{y} = \sum_{j=1}^k \beta_j y_j = \frac{2\pi}{N+1} \sum_{j=1}^k y_j 2^{j-1} = \frac{2\pi y}{N+1}. \tag{10}$$

$$|\Psi\rangle = |y\rangle_C \otimes R_z^{\otimes n}\left(\frac{2\pi y}{N+1}\right) |\psi\rangle_S \otimes \frac{1}{\sqrt{N+1}} \exp\left\{\left(-\frac{i}{2}\Gamma\right)\right\} \sum_{y=0}^N \exp\left\{\left(\frac{in\pi y}{N+1}\right)\right\}. \tag{11}$$

Noting that,

$$R_z^{\otimes n}\left(\frac{2\pi y}{N+1}\right) = \exp\left(\frac{-in\pi y}{N+1}\right) \sum_{x=0}^n \exp\left(\frac{2\pi i x y}{N+1}\right) P_x. \tag{12}$$

after the controlled-Z operation, we see the $|\Psi\rangle$ can be expressed as:

$$|\Psi\rangle = \frac{1}{\sqrt{N+1}} \exp\left\{\left(-\frac{i}{2}\Gamma\right)\right\} \sum_{y=0}^N \sum_{x=0}^n \exp\left(\frac{2\pi i x y}{N+1}\right) P_x |\psi\rangle_S |y\rangle_C. \tag{13}$$

After applying IQFT on control register qubits, we have

$$\frac{1}{N+1} \exp\left\{\left(-\frac{i}{2}\Gamma\right)\right\} \sum_{z,y,x=0}^N \exp\left(\frac{2\pi i (x-z)y}{N+1}\right) |z\rangle_C P_x |\psi\rangle_S. \tag{14}$$

Here the summation,

$$S = \sum_{y=0}^N \exp\left(\frac{2\pi i (x-z)y}{N+1}\right) = (N+1)\delta_{xz} \tag{15}$$

The state finally becomes:

$$\exp\left\{\left(-\frac{i}{2}\Gamma\right)\right\}\sum_{x=0}^N|x\rangle_C\otimes P_x|\psi\rangle_S \quad (16)$$

Therefore, probability that we measure Hamming weight a is:

$$p_a = |P_a|\psi\rangle|^2 \quad (17)$$

and the post-measurement state $|\bar{\psi}\rangle$ is:

$$|\bar{\psi}\rangle = \frac{P_a|\psi\rangle}{\sqrt{p_a}}. \quad (18)$$

Exercise 3

Use the function created in part 2 above to measure the Hamming weight of the following states:

a) $|\psi\rangle = |11\rangle$

The given Hamming weight is 2 the circuit execution results also shows the hamming weight is 2.

b) $|\psi\rangle = |01\rangle$

The given Hamming weight is 1 the circuit execution results also shows the hamming weight is 1.

c) $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

The given state has a probability of $\frac{1}{2}$ to be observed Hamming weight 0, and $\frac{1}{2}$ to be observed Hamming weight 2.

d) $|\psi\rangle = \frac{1}{\sqrt{3}}(|011\rangle + |101\rangle + |110\rangle)$

The given Hamming weight is 2 the circuit execution results also shows the hamming weight is 2.

e) Explain the result in part c) above.

The given state has a probability of $\frac{1}{2}$ to be observed Hamming weight 0, and $\frac{1}{2}$ to be observed Hamming weight 2.

Exercise 4

Explain how the function in part 2 above can be used to prepare a Dicke state.

The post-measurement state can be used to prepare Dicke state. From the Eq. 18, we see that the initially given state projected on to the space of fixed Hamming weight x , therefore, becomes a linear combination computational bases of Hamming weight x . For example of three-qubit system, and $x = 2$, the basis is $\{|011\rangle, |101\rangle, |110\rangle\}$. Here Dicke state in n -qubit system and Hamming weight k has form:

$$|D_{n,k}\rangle = \frac{1}{\sqrt{{}_nC_k}} \sum_{i=1}^{{}_nC_k} |e_i\rangle \quad (19)$$

where $|e_i\rangle$ is an element of basis of Hamming weight k whose the number of basis is ${}_nC_k$. Considering the whole coefficients are equal, the initial state before the measuring Hamming weight also has the same coefficients in the basis of computational basis. Such state is just $+$ state:

$$|\psi\rangle = |++\cdots+\rangle = \frac{1}{\sqrt{2^n}} \sum_{i=0}^{2^n-1} |i\rangle \quad (20)$$

where i denotes the bit string. After measuring Hamming weight k , the post-measurement state $|\bar{\psi}\rangle$ becomes Dicke state $|D_{n,k}\rangle$.

Exercise 5

Suppose we want to prepare the Dicke state with $n = 100$ and $k = 50$.

What is the probability that the state will be measured?

Preparing $+$ state, $|\psi\rangle = |+\rangle^{\otimes n}$, the probability that we measure Hamming weight k is given in the Eq. 17. Among the computational basis, $\{|0\rangle, |1\rangle, \dots, |2^n-1\rangle\}$, the number of basis whose Hamming weight is ${}_nC_k$. Then,

$$p_k = |P_k |\psi\rangle|^2 = \left| \frac{{}_nC_k}{\sqrt{2^n}} \right|^2 = \frac{({}_nC_k)^2}{2^n}. \quad (21)$$

What is the minimum number of measurements required?

How many gates are required?

References

- [1] Soorya Rethinasamy, Margarite L. LaBorde, and Mark M. Wilde. Logarithmic-depth quantum circuits for hamming weight projections, 2024.